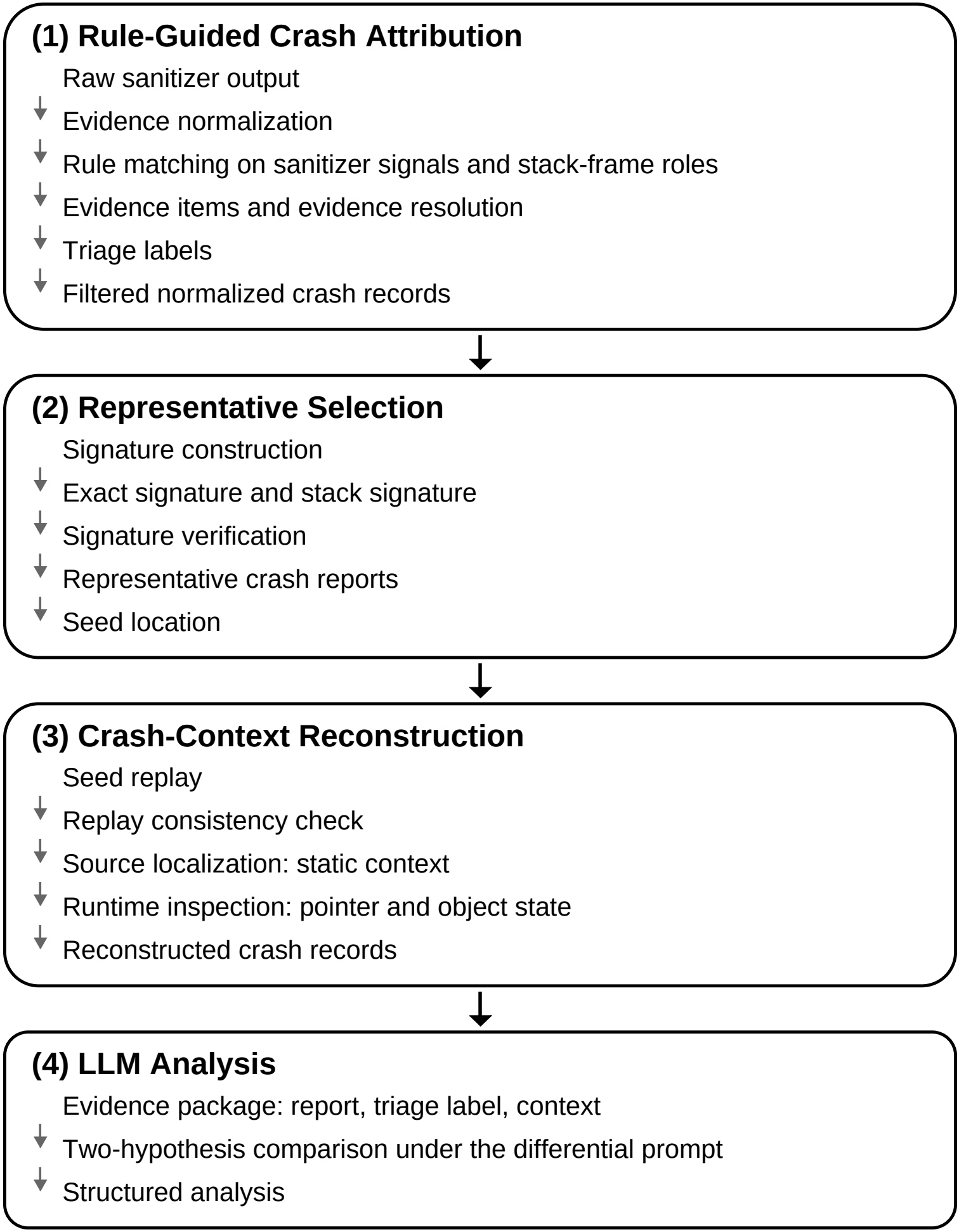
(1) Rule-Guided Crash Attribution
Raw sanitizer output
Evidence normalization
Rule matching on sanitizer signals and stack-frame roles
Evidence items and evidence resolution
Triage labels
Filtered normalized crash records
(2) Representative Selection
Signature construction
Exact signature and stack signature
Signature verification
Representative crash reports
Seed location
(3) Crash-Context Reconstruction
Seed replay
Replay consistency check
Source localization: static context
Runtime inspection: pointer and object state
Reconstructed crash records
(4) LLM Analysis
Evidence package: report, triage label, context
Two-hypothesis comparison under the differential prompt
Structured analysis